

Qiskit Global Summer School 2026

Entanglement & Quantum Effects

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Part 1: What is Entanglement, Really?

Smart people have been confused about this for 100 years

Two ways people get it wrong:

Failure mode 1 (Popular Press):

- Its just spooky magic, don't try to understand it

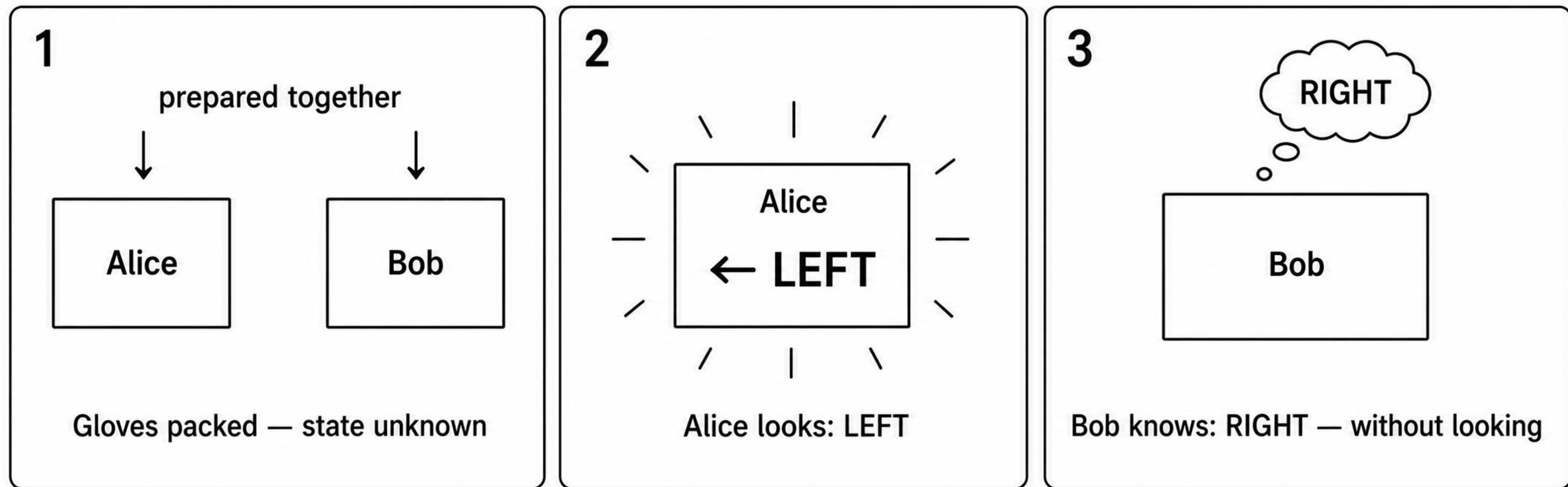
Failure mode 2 (Einstein):

- Its just correlation, boring classical stuff

Today, we do better than both of these.



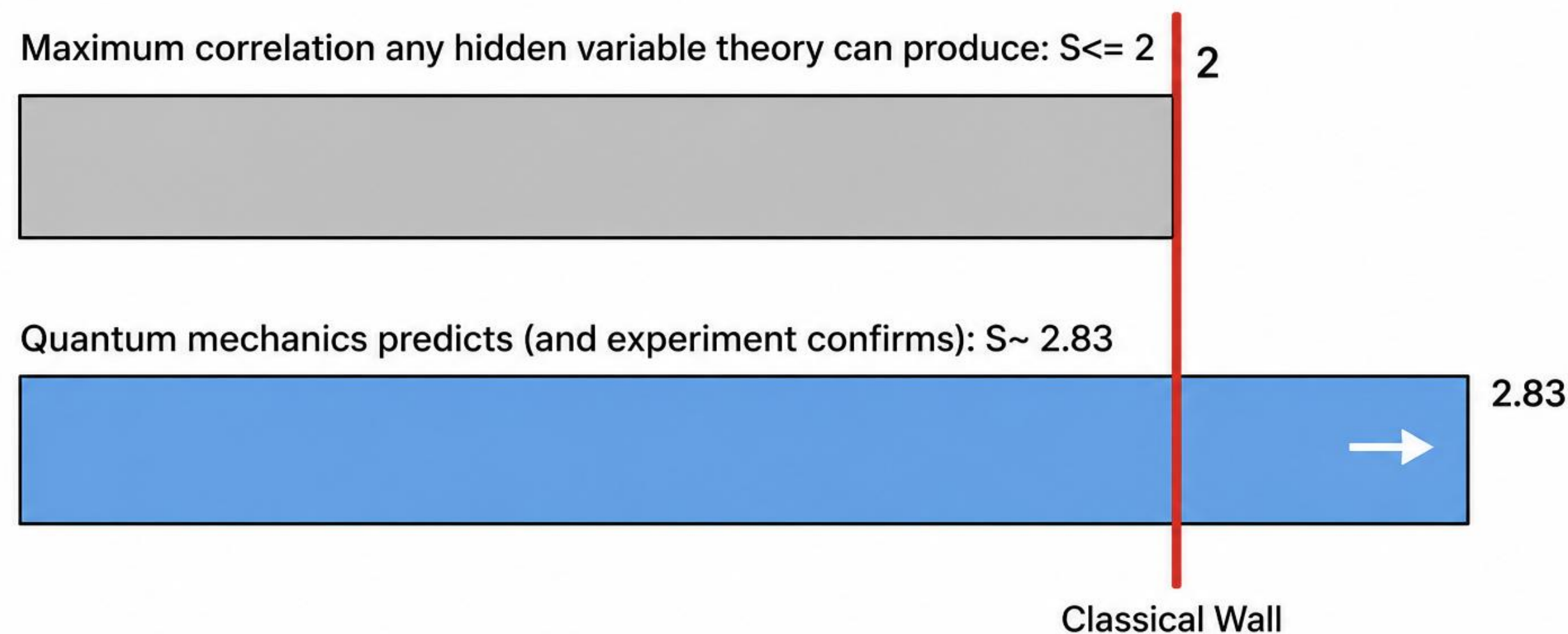
Classical Correlation: The Baseline



CHSH/Bell Tests: Proof It's Not Hidden Variables

If nature were secretly classical (hidden instructions telling each particle what to do) those instructions can only produce correlations up to a certain strength. Quantum mechanics predicts stronger correlations, and experiments confirm them.

Entanglement is not ignorance about a pre-existing state. The CHSH violation is the experimental receipt.



CHSH: Clauser, Horne, Shimony, Holt.

These physicists made Bell's idea experimentally testable.

First confirmed experimentally: Alain Aspect, 1982. Loophole-free tests: 2015.

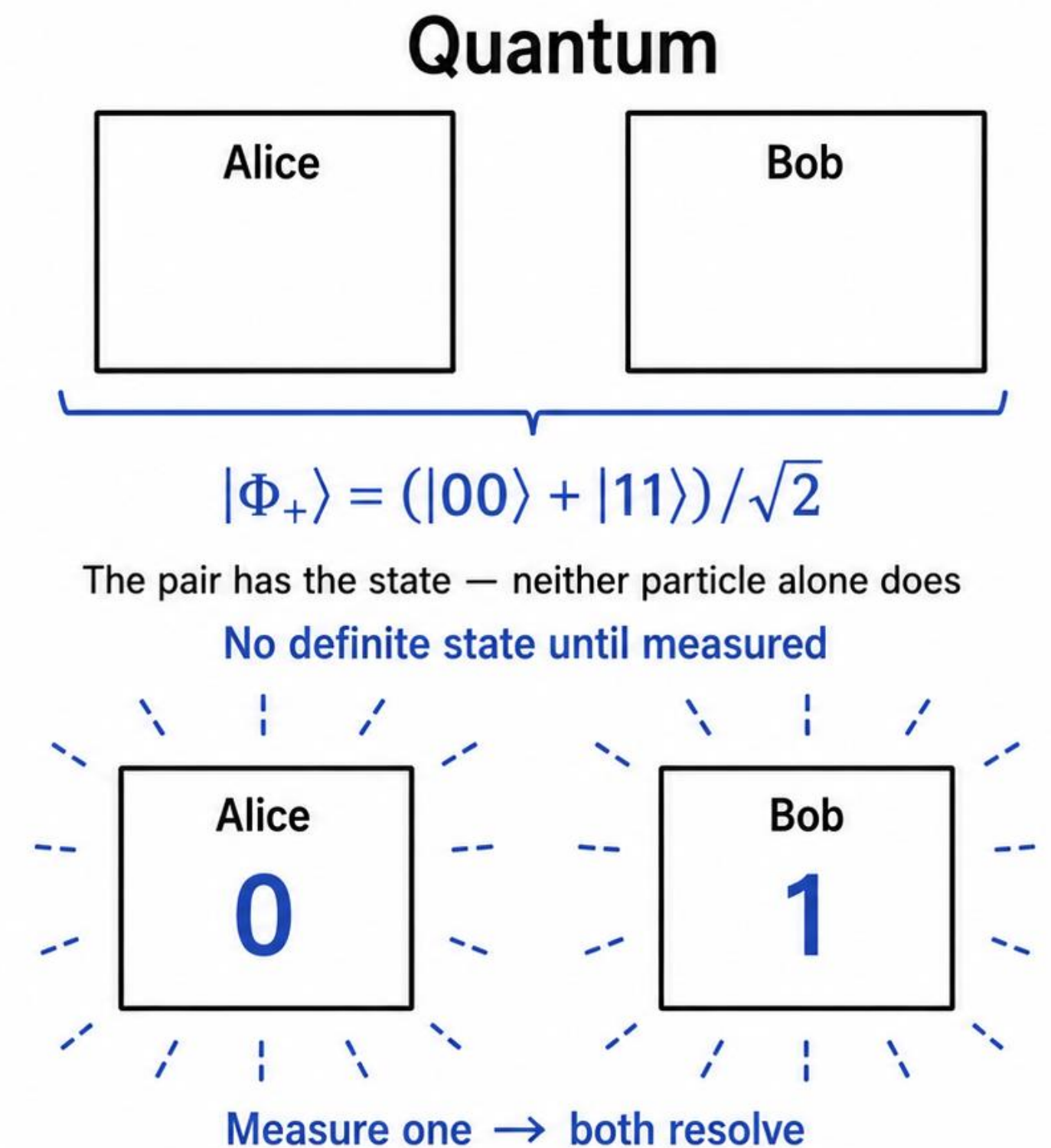
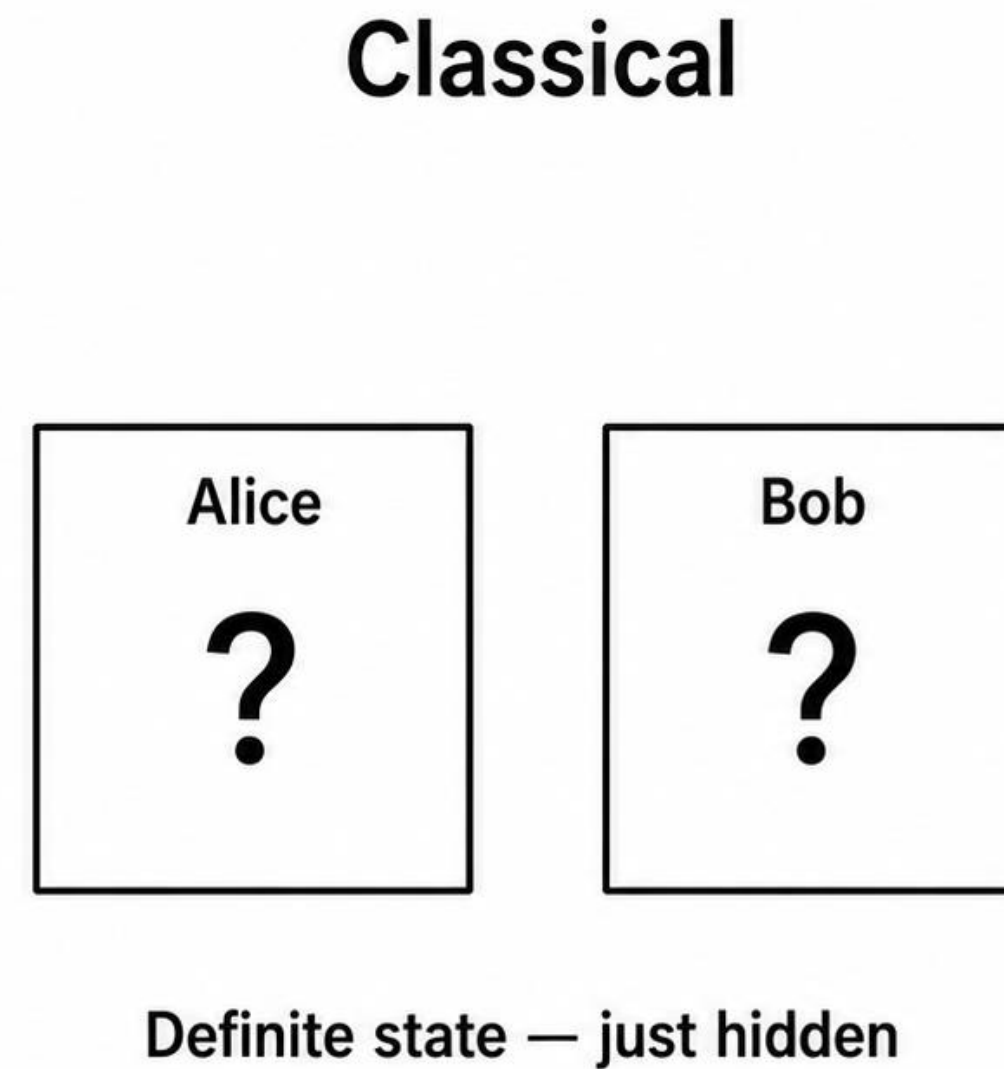
Superposition + Correlation: The Real Difference

- **Classical:** Outcomes are hidden but predetermined
- **Quantum:** Outcomes are genuinely indeterminate until measurement occurs

$$|\Phi_+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

(4 Bell states total)

The joint system has a definite state. The individual particles DON'T.

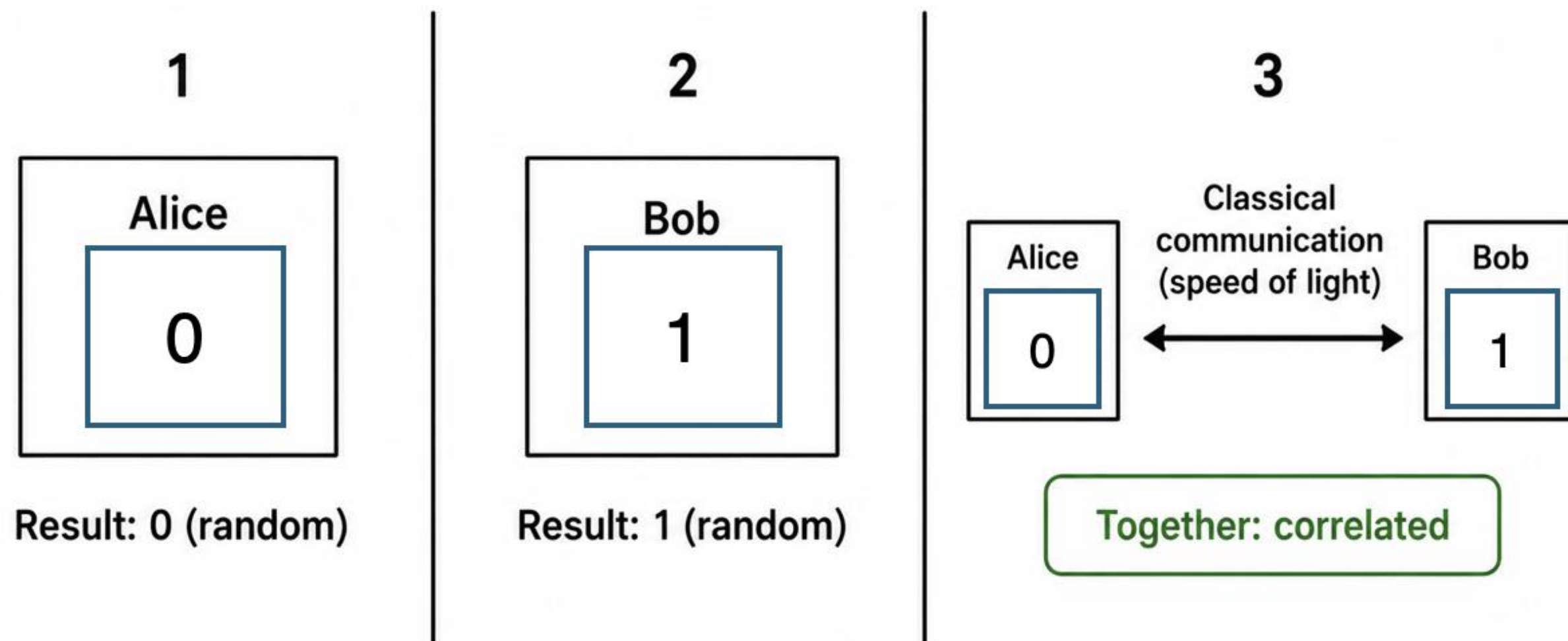


Part 2: What Entanglement Does NOT Mean

Measurement non-locality: the right framing

- ✓ Measuring one particle *instantly* affects what you'll find at the other
- ✗ You cannot use this to send information

Each side sees random outcomes individually. The correlation only appears when Alice and Bob compare notes, and comparing notes requires classical communication. This step is limited by the speed of light.



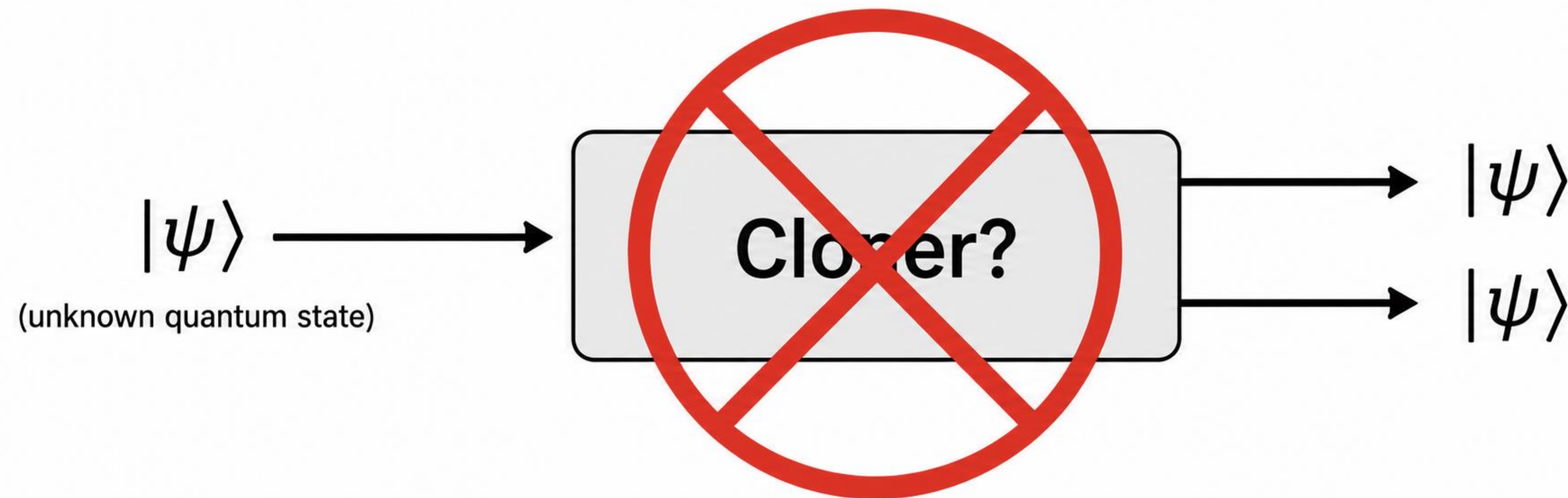
- The non-locality is in the correlations, not the signals
- Einstein's "spooky action" is real, but it carries no message

No-cloning theorem

To copy $\rightarrow$ must know the state
To know the state $\rightarrow$ must measure it
To measure $\rightarrow$ disturbs it
 $\therefore$ The copy destroys the original

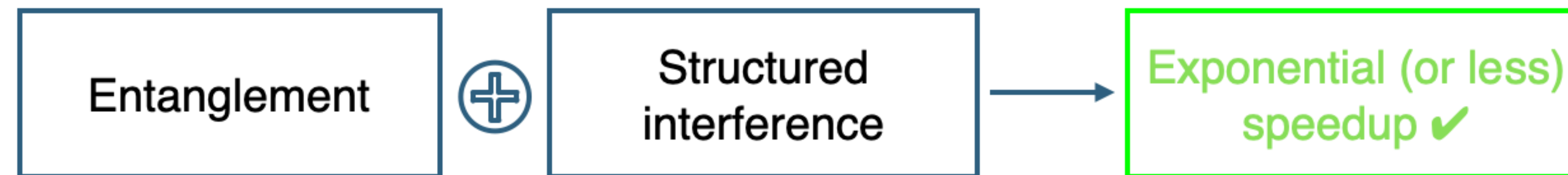
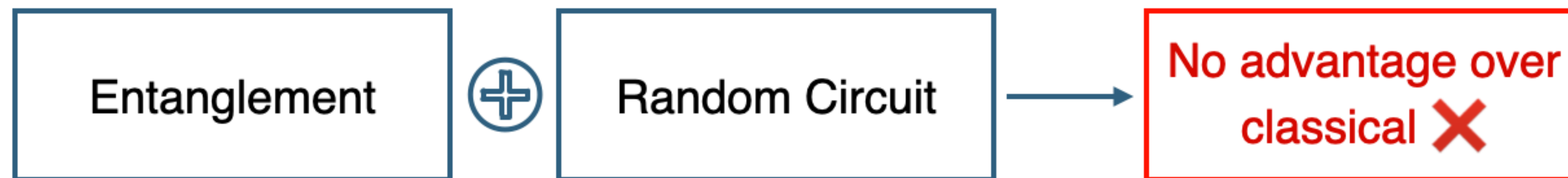
Implications:

Error correction: Encode, don't copy
Cryptography: Eavesdropping always leaves traces



Unknown quantum states cannot be copied — no matter the machine.

Entanglement is a resource, like a powerful ingredient when used in the right recipe.



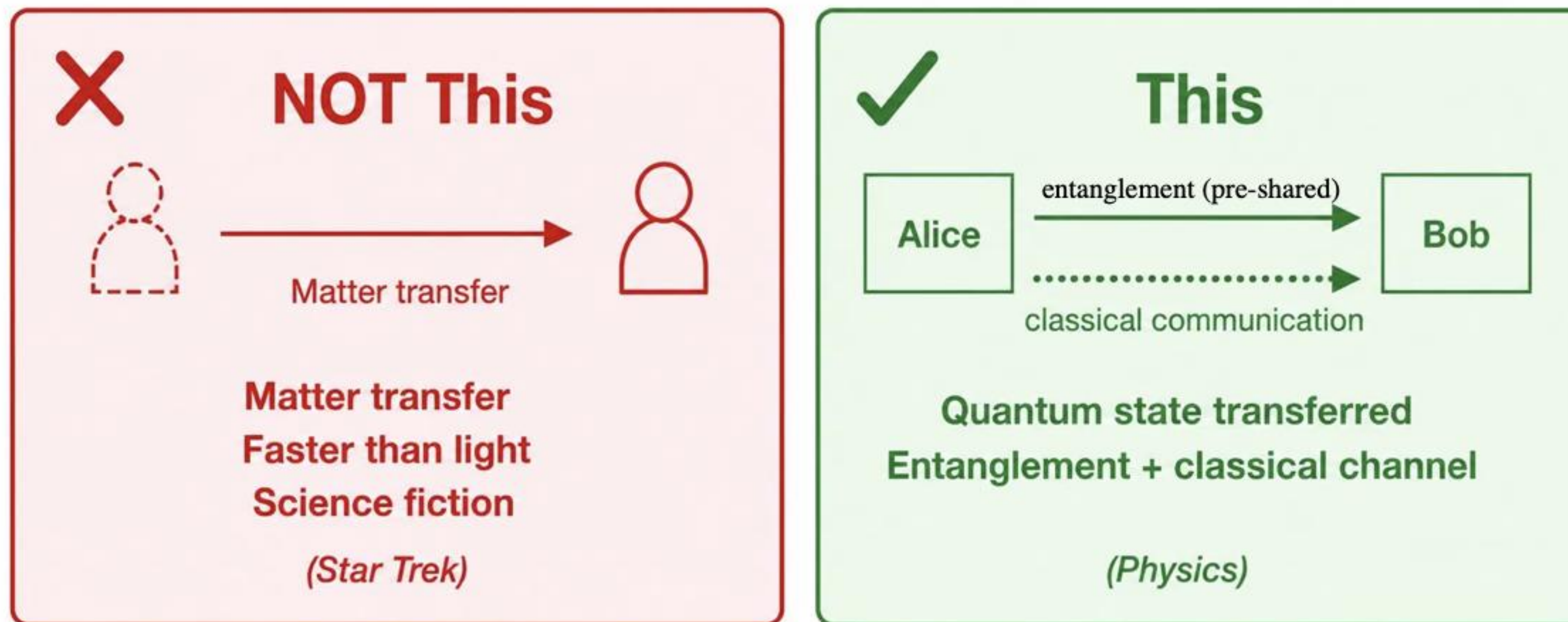
(e.g. Shor's, Grover's)

The algorithm does the work and entanglement enables it

Part 3: What Entanglement Enables: Teleportation as the Worked Example

Quantum teleportation setup

Goal: Transfer an unknown quantum state from Alice to Bob using a pre-shared Bell pair + classical communication



Entanglement is the resource. Classical communication is the activation key.

The three-step protocol

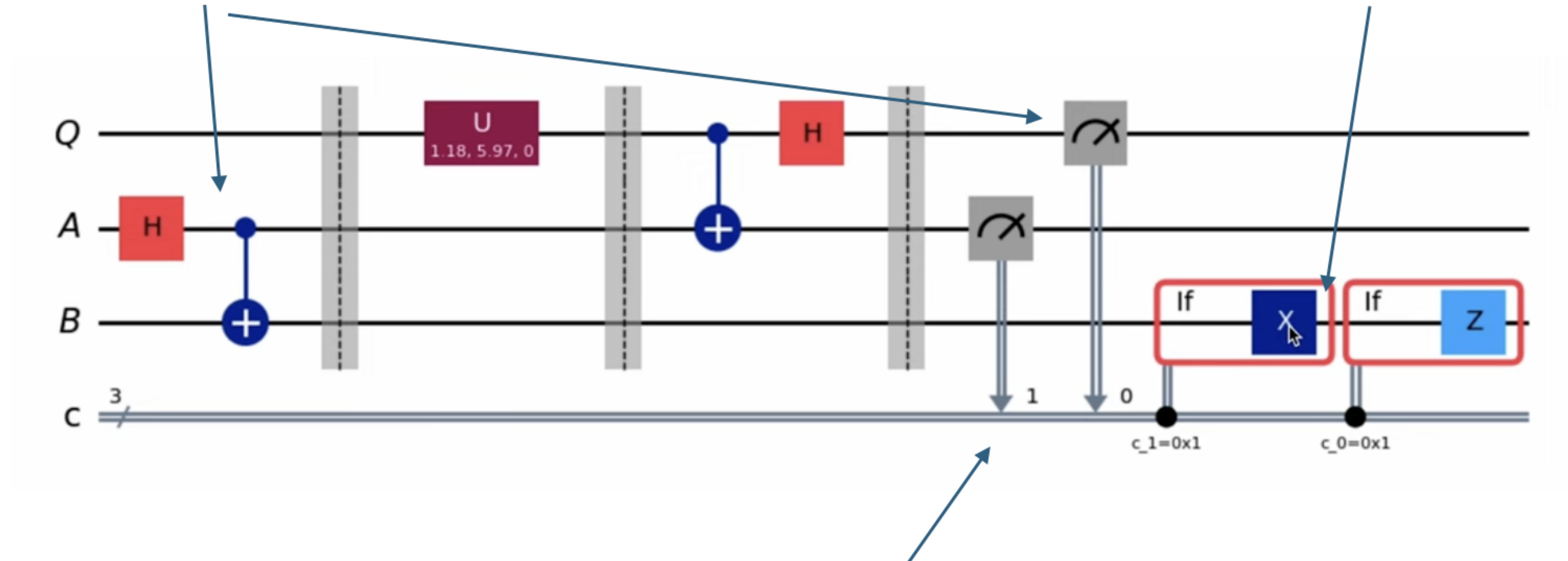
1. Alice applies CNOT + Hadamard to her qubits, then measures both → 2 classical bits
2. Alice sends those 2 bits to Bob (classical channel, the speed-of-light limited step)
3. Bob applies X and/or Z corrections based on those bits to recover the original state exactly

The entanglement does the heavy lifting *before* the classical message; the classical message *activates* it

The three-step protocol: Circuit

1) Alice's operations: CNOT + H + measure

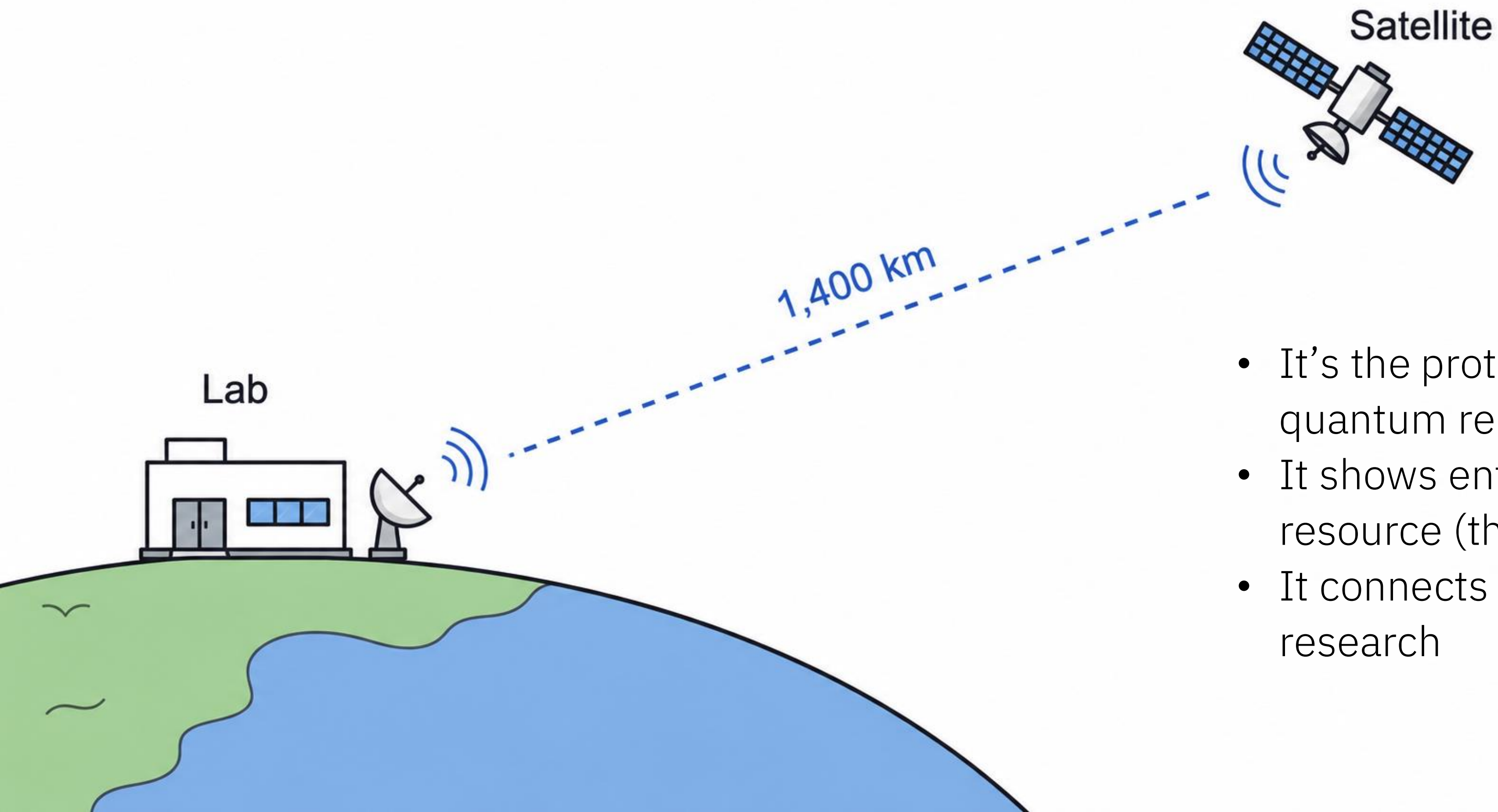
3) Bob's conditional corrections: If X, If Z



2) Classical bits travel to Bob (C register)

Quantum teleportation matters beyond the demo

Quantum teleportation was first proposed in 1993. States have since been teleported over 1,400 km from a lab to a satellite.



- It's the prototype for quantum networking and quantum repeaters
- It shows entanglement is a consumable quantum resource (the Bell pair is used up)
- It connects to near-term quantum internet research

Hands-On Block

Lab 1 — CHSH Inequality

Sweep measurement angle θ from 0 to 2π

Watch S_1 and S_2 exceed the classical bound of 2

Real hardware: IBM quantum processor

Lab 2 — Quantum Teleportation

Run the full 3-step protocol in Qiskit

Verify: Bob's qubit returns to $|0\rangle$ after inverse unitary

Real hardware: IBM quantum processor

Closing

Entanglement is real and experimentally verified. It's not magic.

It doesn't allow faster-than-light anything, but it does allow for non-classical correlations

It's a resource useful only with the right algorithm or protocol

Teleportation is proof-of-concept for what entanglement really enables.